

Lec 25

- ① Single Function of a RV
- ② One Function of two R.Vs.
- ③ Two R.Vs that are functions of two R.Vs.

$$\longrightarrow Y = g(X)$$

① Find CDF $F_Y(y)$

② Differentiate $F_Y(y)$ to get $f_Y(y)$.

Example: ① $Y = \sqrt{X}$

$$F_Y(y) = F_X(y^2)$$

$$f_Y(y) = f_X(y^2) 2y$$

$$f_Y(y) = \frac{\partial F_X(y^2)}{\partial y}$$

$$= f_X(y^2) 2y$$

(2) $Y = X^2$ where $f_X(x)$ is known. Find $f_Y(y)$.

$$\begin{aligned} F_Y(y) &= P(Y \leq y) \\ &= P(X^2 \leq y) \\ &= P(-\sqrt{y} \leq X \leq \sqrt{y}) \\ &= F_X(\sqrt{y}) - F_X(-\sqrt{y}) \end{aligned}$$

$$\begin{aligned} f_Y(y) &= \frac{\partial F_Y(y)}{\partial y} \\ &= \frac{\partial F_X(\sqrt{y})}{\partial y} - \frac{\partial F_X(-\sqrt{y})}{\partial y} \end{aligned}$$

X continuous

Example: $X \sim U$

$$Y|X=x$$

$$X|Y=y$$

$$= f_X(\sqrt{y}) \frac{1}{2} \frac{1}{\sqrt{y}}$$

$$= \frac{1}{2\sqrt{y}} [f_X(\sqrt{y}) + f_X(-\sqrt{y})]$$

x^2 where $f_X(x)$ is
on find $f_Y(y)$.

$$= P(Y \leq y)$$

$$= P(x^2 \leq y)$$

$$P(-\sqrt{y} \leq x \leq \sqrt{y})$$

$$F_X(\sqrt{y}) - F_X(-\sqrt{y})$$

$$F_X(y)$$

$$\frac{\partial}{\partial y}$$

$$\frac{\partial F_X(\sqrt{y})}{\partial y} - \frac{\partial F_X(-\sqrt{y})}{\partial y}$$

X continuous, Y discrete

Example: $X \sim \text{Uniform}[0,1]$

$$Y|X=x \sim \text{Bernoulli}(x)$$

$$X|Y=y$$

$$\rightarrow Y|X=x \sim \text{Binomial}(n, x)$$

$$X \sim \text{Beta}(\alpha, \beta)$$

$$f_X(x) = \frac{x^{\alpha-1} (1-x)^{\beta-1}}{B(\alpha, \beta)} \quad x \in [0,1]$$

$$\text{where } B(\alpha, \beta) = \int_0^1 x^{\alpha-1} (1-x)^{\beta-1} dx$$

$$= f_X(\sqrt{y}) \cdot \frac{1}{2} y^{1/2} - f_X(-\sqrt{y}) \cdot \left(-\frac{1}{2} y^{1/2}\right)$$

$$= \frac{1}{2\sqrt{y}} [f_X(\sqrt{y}) + f_X(-\sqrt{y})]$$

$\in [0, 1]$

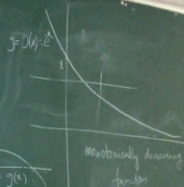
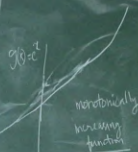
we

$$p = \int_0^1 x^{\alpha-1} (1-x)^{\beta-1} dx$$

Shift and scale of a R.V

$$Y = aX + b$$

$$f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right)$$



Let $Y = g(X)$ where
function g is monotonic in interval

I where $f_X(x)$ is non-zero.

increasing or
decreasing

$$g(h(y)) = y$$

Let h be a function st $h(y) = x$
 $y = g(x)$

$$f_Y(y) = \frac{f_X(x)}{g'(x)} = \frac{f_X(h(y))}{g'(h(y))} \quad \text{where } x = h(y)$$

$$F_Y(y) = P(Y \leq y) = P(g(X) \leq y)$$

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$$F_Y(y) = P(g(X) \leq y)$$

$$= \begin{cases} P(X \leq h(y)) \end{cases}$$

$g(x)$ is increasing.

$$\begin{cases} P(X \geq h(y)) \end{cases}$$

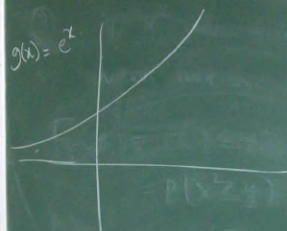
$g(x)$ is decreasing.

$$= \begin{cases} F_X(h(y)) & g(x) \text{ is } \uparrow \end{cases}$$

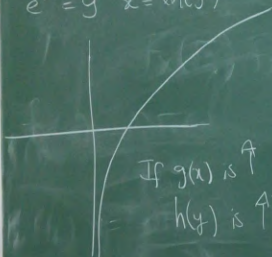
$$\begin{cases} 1 - F_X(h(y)) & g(x) \text{ is } \downarrow \end{cases}$$

$f_Y(y)$

$$= \frac{\partial F_Y(y)}{\partial y} = \begin{cases} f_X(h(y)) h'(y) \\ -f_X(h(y)) h'(y) \end{cases} = f_X(h(y)) |h'(y)|$$



$$e^x = y \quad x = \ln(y)$$



If $g(x)$ is $\uparrow$
 $h(y)$ is $\uparrow$

$$g(h(y)) = y$$

$$\frac{\partial g(h(y))}{\partial y} = 1$$

$$g'(h(y)) h'(y) = 1$$

$$f_Y(y) = f_X(h(y)) |h'(y)|$$

$$= \frac{f_X(h(y))}{|g'(h(y))|}$$

$$= \frac{f_X(x)}{|g'(x)|}$$

②

Check example using this formula

$$\textcircled{1} Y = \sqrt{x}$$

$$g'(x) = \frac{1}{2\sqrt{x}}$$

$$\begin{aligned} f_Y(y) &= \frac{f_X(\bar{x})}{\frac{1}{2\sqrt{x}}} = 2\sqrt{x} f_X(x) \\ &= 2y f_X(y^2). \end{aligned}$$

② Single function of two R.V.s.

Examples (a) $Z = X + Y$

Given $f_{X,Y}(x,y)$ find $f_Z(z)$.

$$= \frac{1}{2\sqrt{x}}$$

$$f_X(x)$$

$$F_Z(z) = P(X+Y \leq z)$$

$$= \iint_{(x,y): x+y \leq z} f_{X,Y}(x,y) dx dy$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{z-y} f_{X,Y}(x,y) dx dy$$

Leibniz rule for differentiation

$$G(z) = \int_{a(z)}^{b(z)} f(z,t) dt$$

$$\frac{\partial G(z)}{\partial z} = f(z, b(z)) \frac{\partial b(z)}{\partial z} - f(z, a(z)) \frac{\partial a(z)}{\partial z} + \int_{a(z)}^{b(z)} \frac{\partial f(z,t)}{\partial z} dt$$

$$\frac{\partial F_Z(z)}{\partial z} = \int_{-\infty}^{\infty} \frac{\partial}{\partial z} \left[\int_{-\infty}^{z-y} f_{X,Y}(x,y) dx \right] dy$$

$$= \int_{-\infty}^{\infty} \left[f_{X,Y}(z-y, y) \times 1 \right] + 0 dy$$

$$f_Z(z) = \int_{-\infty}^{\infty} f_{X,Y}(z-y, y) dy$$

If X, Y are non-negative $f_Z(z) = \int_0^z f_{X,Y}(z-y, y) dy$

If Y is non-negative $f_Z(z) = \int_0^{\infty} f_{X,Y}(z-y, y) dy$

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Q.1 $Z = X + Y$

if X, Y are independent

increasing

$$f_Z(z) = \int_{-\infty}^{\infty} f_{X,Y}(z-y, y) dy$$

$$= \int_{-\infty}^{\infty} f_X(z-y) f_Y(y) dy$$

② $Z = X/Y$

Joint density of X, Y is given

find pdf of Z

$$F_Z(z) = P\left(\frac{X}{Y} \leq z\right) = P(Y \geq 0, X \leq Yz) + P(Y \leq 0, X \geq Yz)$$

Exercise

One function of one R.V.

$$y = g(x) \text{ and } x = h(y)$$

$$f_Y(y) = \frac{f_X(x)}{g'(x)} = f_X(h(y)) h'(y).$$

$$\begin{aligned} P(y \leq Y \leq y+dy) &= f_Y(y) dy \\ &= P(x \leq X \leq x+dx) \\ &= f_X(x) dx. \end{aligned}$$



$$f_Y(y) = \frac{f_X(x)}{\frac{dy}{dx}} = f_X(x) \frac{dx}{dy}$$

Suppose there
is $g(x)$.

Example $g(x)$
 $x_1 =$

$$\begin{aligned} P(y \leq Y \leq y+dy) &= P(x_1 \leq X \leq x_1+dx) \\ &+ P(x_2 \leq X \leq x_2+dx) \end{aligned}$$

$$f_Y(y) =$$

$$e = h(y)$$

$$f_X(h(y)) h'(y).$$

$$= f_Y(y) dy$$

$$(\leq x + dx)$$

$$dx$$

$$f_Y(y) = \frac{f_X(x)}{\frac{dy}{dx}} = f_X(x) \frac{dx}{dy}$$

Suppose there are $x_1, x_2, \dots, x_n$

$$\text{such that } g(x_i) = y$$

Example $g(x) = x^2$

$$x_1 = \sqrt{y}, x_2 = -\sqrt{y}$$

$$P(y \leq Y \leq y + dy)$$

$$= P(x_1 \leq X \leq x_1 + dx_1)$$

$$+ P(x_2 \leq X \leq x_2 + dx_2)$$

$$f_Y(y) = \frac{f_X(x_1) \left| \frac{dx_1}{dy} \right|}{\left| \frac{dy}{dx} \right|} + \frac{f_X(x_2) \left| \frac{dx_2}{dy} \right|}{\left| \frac{dy}{dx} \right|}$$

$$= \frac{f_X(\sqrt{y})}{2\sqrt{y}} + \frac{f_X(-\sqrt{y})}{2\sqrt{y}}$$

Two functions of two R.V.s

$$Z = g_1(X, Y)$$

$$W = g_2(X, Y)$$

Find Joint PDF $f_{Z,W}(z,w)$

given $f_{X,Y}(x,y)$

$$P((Z,W) \in \Delta_{z,w})$$

Box interval around z,w

$$= P((X,Y) \in \Delta_{x,y})$$

(x,y) st

$$Z = g_1(x,y)$$

$$W = g_2(x,y)$$

$$(x,y) \iff (z,w)$$

$$f_{Z,W}(z,w) dz dw$$

$$= f_{X,Y}(x,y) dx dy$$

$$f_{Z,W}(z,w) = \frac{f_{X,Y}(x,y)}{\left| \frac{\partial z}{\partial x} \frac{\partial w}{\partial y} - \frac{\partial z}{\partial y} \frac{\partial w}{\partial x} \right|}$$

$$\left| \frac{\partial z}{\partial x} \frac{\partial w}{\partial y} - \frac{\partial z}{\partial y} \frac{\partial w}{\partial x} \right|$$

Jacobian (z,w)
w.r.t (x,y)

//

$$\left| \frac{dz}{dx} \frac{dw}{dy} - \frac{dz}{dy} \frac{dw}{dx} \right|$$

$$\left| \frac{dw}{dx} \frac{dz}{dy} - \frac{dw}{dy} \frac{dz}{dx} \right|$$



Two functions of two R.V.s

$$Z = g_1(X, Y)$$

$$W = g_2(X, Y)$$

Find Joint PDF $f_{Z,W}(z,w)$

given $f_{X,Y}(x,y)$

$$P((Z,W) \in \Delta_{z,w})$$

Box interval around z,w

$$= P((X,Y) \in \Delta_{x,y})$$

$$\begin{aligned} f_{Z,W}(z,w) dz dw \\ = f_{X,Y}(x,y) dx dy \end{aligned}$$

$$f_{Z,W}(z,w) = \frac{f_{X,Y}(x,y)}{\boxed{\frac{\partial Z}{\partial x} \frac{\partial W}{\partial y}}}$$

(x,y) st

$$Z = g_1(x,y)$$

$$W = g_2(x,y)$$

$$(x,y) \iff (z,w)$$

$$f_{z,w}(z,w) dz dw = f_{x,y}(x,y) dx dy$$

$$f_{z,w}(z,w) = \frac{f_{x,y}(x,y)}{\boxed{\frac{dz dw}{dx dy}}}$$

$$\boxed{\frac{dz dw}{dx dy}}$$

Jacobian (z,w)
 x,y

//

$$\begin{vmatrix} \frac{dz}{dx} & \frac{dz}{dy} \\ \frac{dw}{dx} & \frac{dw}{dy} \end{vmatrix}$$

(x,y) st

$$z = g_1(x,y)$$

$$w = g_2(x,y)$$

$$(x,y) \iff (z,w)$$

$$f_{z,w}(z,w) =$$

x,y st

$$z = g_1(x,y)$$

$$w = g_2(x,y)$$

Examples:

Find joint

$$R = \sqrt{x^2 + y^2}$$

$$x = r \cos \theta$$

$$f_{R,\theta}(r,\theta)$$

$$f_{z,w}(z,w) = f_{x,y}(x,y)$$

$$\begin{aligned} x, y \text{ st} \\ z = g_1(x,y) \\ w = g_2(x,y) \end{aligned} \quad \begin{aligned} & \frac{J_{x,y}(z,w)}{J_{x,y}(x,y)} \\ & = f_{x,y}(x,y) \frac{J_{x,y}(x,y)}{J_{x,y}(z,w)} \end{aligned}$$

$$\frac{d}{dy} \left(\frac{z}{y} \right)$$

$$\frac{dw}{dy}$$

Examples: X, Y are $N(0,1)$, independent

Find joint pdf of R, Θ

$$R = \sqrt{X^2 + Y^2}, \quad \Theta = \tan^{-1}\left(\frac{Y}{X}\right)$$

$$x = r \cos \theta, \quad y = r \sin \theta$$

$$f_{R,\Theta}(r,\theta) = f_{x,y}(r \cos \theta, r \sin \theta) \cdot r$$

$$X \sim N(0,1)$$

$$f_X(x) = \frac{e^{-x^2/2}}{\sqrt{2\pi}}$$

$$f_{X,Y}(x,y) = \frac{e^{-x^2/2 - y^2/2}}{2\pi}$$

$$= \frac{e^{-r^2/2}}{2\pi}$$

r